

Generation Interconnection Request System Impact Study Report

For Queue Project Q14-0372

Medora 345 kV
October 24, 2016

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q14-0372. The Interconnection Customer proposes a 306.3 MW wind facility interconnecting at the Medora 345 kV substation in the Pennsylvania area. The request was received on August 30, 2014. The study evaluated system intact and single contingency conditions using the regional planning case, including 153 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 9 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$8,433,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Lakehead Naubinway - Medora 138 KV
Contingency: P.T.S - Medora 345 KV
Loading: 146.4% (Rating: 269 MVA)
DFAX: 7.2%
Pre-project loading: 138.3%
2. Facility: Lakehead Naubinway - Spring Vly 138 KV
Contingency: Medora - P.T.S Substation 345 KV
Loading: 143.0% (Rating: 261 MVA)
DFAX: 7.2%
Pre-project loading: 134.6%
3. Facility: Sara-2 - Spring Vly 138 KV
Contingency: Medora - P.T.S 345 KV
Loading: 139.8% (Rating: 255 MVA)
DFAX: 7.2%
Pre-project loading: 131.2%
4. Facility: Sara-2 - Manati 8401 138 KV
Contingency: Medora - P.T.S 345 KV Line
Loading: 139.7% (Rating: 248 MVA)
DFAX: 7.2%
Pre-project loading: 130.8%
5. Facility: Parleys - Manati 8401 138 KV
Contingency: Medora - P.T.S Substation 345 KV
Loading: 135.0% (Rating: 204 MVA)
DFAX: 5.2%
Pre-project loading: 127.2%

6. Facility: Parleys - Gregg 138 KV L
 Contingency: Medora - P.T.S 345 KV
 Loading: 109.4% (Rating: 229 MVA)
 DFAX: 5.2%
 Pre-project loading: 102.5%
7. Facility: P.T.S Substation - Attica 138 KV
 Contingency: Somerville - P.T.S 345 KV
 Loading: 108.8% (Rating: 260 MVA)
 DFAX: 5.1%
 Pre-project loading: 102.8%
8. Facility: Edes Tap 2 - West Campus 345 KV
 Contingency: Orchard Rd Substation - Shawville Station 500 KV Line
 Loading: 103.3% (Rating: 1,155 MVA)
 DFAX: 5.1%
 Pre-project loading: 101.9%
9. Facility: Gregg - Dravosburg 138 KV line
 Contingency: Medora - P.T.S Substation 345 KV
 Loading: 102.4% (Rating: 237 MVA)
 DFAX: 5.2%
 Pre-project loading: 95.7%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Lakehead Naubinway - Medora 138 KV	Rebuild line with higher-capacity conductor	\$3,065,000	23% (\$700,000)
Lakehead Naubinway - Spring Vly 138 KV	Rebuild line with higher-capacity conductor	\$1,071,000	23% (\$245,000)
Sara-2 - Spring Vly 138 KV	Rebuild line with higher-capacity conductor	\$5,728,000	23% (\$1,308,000)
Sara-2 - Manati 8401 138 KV	Rebuild line with higher-capacity conductor	\$976,000	23% (\$223,000)
Parleys - Manati 8401 138 KV	Rebuild line with higher-capacity conductor	\$2,338,000	100% (\$2,338,000)
Parleys - Gregg 138 KV L	Reconductor line	\$461,000	100% (\$461,000)
P.T.S Substation - Attica 138 KV	Reconductor line	\$1,075,000	100% (\$1,075,000)
West Camps - Edes Tap 2 345 KV L	Reconductor line	\$13,006,000	7% (\$908,000)
Gregg - Dravosburg 138 KV	Reconductor line	\$1,175,000	100% (\$1,175,000)
	Total allocated to project		\$8,433,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.